

# What Can Money Buy?

## Inequality and Fiscal Policy Implications of Citizens United v. Federal Election Commission

Alessio Rebechi<sup>1</sup> · Philippe Van Kerm<sup>2,3</sup> · Piotr Paradowski<sup>3</sup> · Anthony Lepinteur<sup>2</sup> · Nicholas Rohde<sup>4</sup>

<sup>1</sup>University of Tasmania · <sup>2</sup>University of Luxembourg · <sup>3</sup>Luxembourg Income Study · <sup>4</sup>Griffith University

IARIW General Conference · Brussels · 28 August 2026

Session: Recent laws and their inequality implications

---

Presented by the discussant: **Max Ghenis** (PolicyEngine)

Author response: Philippe Van Kerm

# Does money in politics change actual policy?

The political effects of Citizens United are documented; the policy effects are not

The paper opens with OpenSecrets' count: in the last presidential election the 100 largest donors gave **\$2.4 billion**.

**Citizens United v. FEC** (January 2010) and **SpeechNow.org v. FEC** (March 2010) removed limits on independent political spending by corporations, unions, and individuals.

A decade of research finds **political** effects: higher Republican vote shares, more conservative state legislatures (Abdul-Razzak et al. 2020; Klumpp et al. 2016; Harvey & Mattia 2022).

"What this implies for policy decisions, however, is unclear. Most research to date has found no impact on actual policy."

— the paper's conclusion. Its claim: this paper supplies the missing link, from campaign finance deregulation to tax schedules to inequality. (We come back to that sentence in the discussion.)



# Sixty years of limits, undone in two rulings

US campaign finance regulation, 1947–2022 (paper Figure 1)





# The natural experiment: 23 states had their own bans

Citizens United voided all of them; states rewrote their laws during 2010 — differentially

## Corporate *and* union ban — 13 states in sample

AK, AZ, MI, NH, NC, ND, OH, OK, PA, RI, TX, WI, WY

## Corporate ban only — 8 states

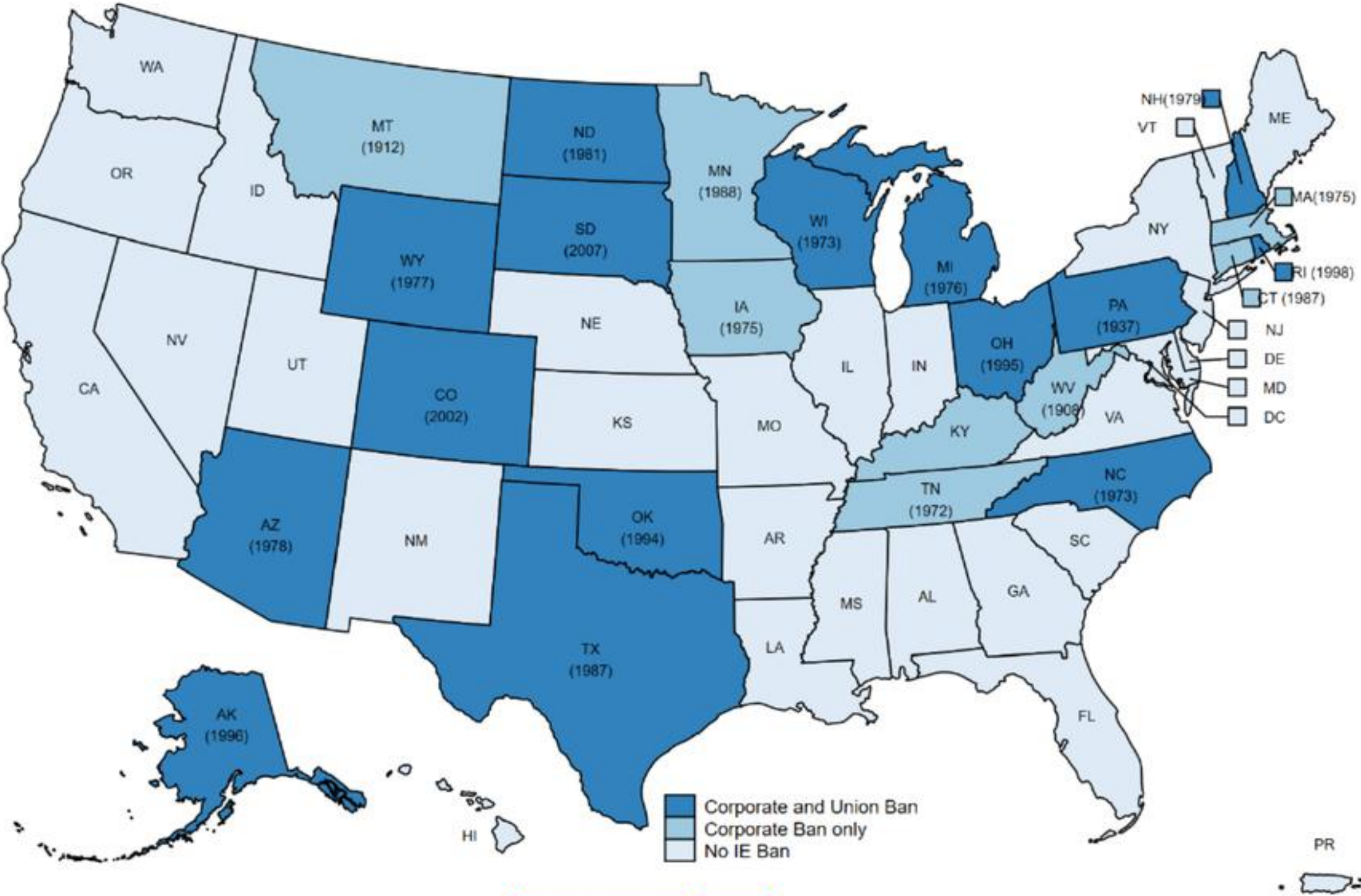
CT, IA, KY, MA, MN, MT, TN, WV

## No ban — controls

The remaining states never restricted independent expenditures

Laws revised to comply between January and October 2010 (Table 2; New Hampshire has no date, and Montana’s ban was back in force from December 2011 to June 2012): a sharp, court-imposed, differential deregulation. CO and SD (bans adopted only in 2002 and 2007) are excluded.

Figure 2: State Campaign Regulations before Citizens United



Notes: The map has been reproduced from Klumpp et al. (2016). Years in parentheses indicate the years of introduction of the funding restrictions.



# What followed: outside spending went from under \$100M to over \$4B a cycle

Paper Table 1 (OpenSecrets): federal outside spending by recipient party, \$ millions — the full series, both sides summed

|                                        | 2004 | 2008 | 2012 | 2016 | 2020               | 2024  |
|----------------------------------------|------|------|------|------|--------------------|-------|
| For Democrats                          | 60   | 121  | 73   | 150  | 644                | 1,062 |
| For Republicans                        | 23   | 20   | 179  | 343  | 961 <sup>†</sup>   | 763   |
| Against Democrats                      | 4    | 34   | 472  | 392  | 1,003              | 1,411 |
| Against Republicans                    | 7    | 32   | 295  | 504  | 961                | 923   |
| Republican-aligned (for R + against D) | 27   | 54   | 651  | 735  | 1,964 <sup>†</sup> | 2,174 |
| Democratic-aligned (for D + against R) | 67   | 153  | 368  | 654  | 1,605              | 1,985 |

The paper’s headline — spending *against* Democrats rose 353-fold — rides on a \$4M 2004 base. Both sides grew; since 2016 the two sides have been within about 10% of each other.

And it went dark: the share of Super PAC spending with **full donor disclosure** fell from 77% (2010) to 33% (2024) (paper Table 3).

<sup>†</sup> As printed, the 2020 column sums to \$3.6B, above OpenSecrets’ 2020 total (\$2.9B excluding party committees; \$3.3B all groups), and “for Republicans” repeats the “against Republicans” figure — likely a transcription slip.



# This paper

From vote shares to tax schedules to inequality

## RESEARCH QUESTION

Did lifting bans on independent political expenditures change the size and progressivity of state income taxes — and post-tax income inequality?

Closest prior work (Slattery, Tazhitdinova & Robinson 2023, *J. Public Economics*) studies statutory rates and aggregate revenues — the tax system *in the books*.

This paper measures the tax system **as households experience it**: average rates by income and wealth group, and the redistribution the schedule actually delivers.

Preview: where states had banned **both corporate and union** spending, top-group tax rates and redistribution fell after the ruling. Where only corporations had been banned, they did not.



# Measurement: 900 simulated tax systems, one fixed population

SCF microdata through NBER TAXSIM — only the law varies

**Survey of Consumer Finances 2011** (LWS-harmonized): the household survey that oversamples the wealthy — credible top-5% and top-1% incomes *and net worth*.

**NBER TAXSIM**: federal and state income tax law for all 50 states, 2004–2021.

The same households are run through **every state-year schedule** (incomes CPI-deflated to each year):

**50 states × 18 years = 900 datasets**

## Why this is clean

With the population held fixed, no composition change, migration, or behavioral response contaminates the measure. What moves is **tax legislation itself** — a pure *de jure* policy panel.

Main text: state taxes assessed on pre-federal income (AGI). Appendix D repeats everything on after-federal-tax income — results are larger and sharper there.

(The SCF is triennial: “2011” is the 2010 wave, LWS US10, with incomes for 2009 — see the replication slide.)



# Summarizing schedules, not statutory parameters

Top marginal rates miss deductions, exemptions, and the base — these indicators capture the whole shape

| Indicator                 | Definition                                                                        | Captures                 |
|---------------------------|-----------------------------------------------------------------------------------|--------------------------|
| Average tax rate (ATR)    | Total state income tax over total pre-tax income                                  | Size of the tax          |
| ATR, top 5% and top 1%    | By income — and, using SCF strengths, by net worth                                | Burden at the top        |
| Progressivity ( $\beta$ ) | ATR of top 20% minus ATR of bottom 20% — defined (eq. 5) but reported in no table | Shape of the schedule    |
| Reynolds–Smolensky index  | Gini(pre-tax) – Gini(post-tax)                                                    | Redistribution delivered |

Each indicator  $\Phi$  is evaluated on the fixed household sample for every state-year law — turning 900 simulated tax systems into a 900-cell panel of schedule characteristics.



# Identification: three complementary designs

"46 states", 2004–2021, N = 846 (= 47 units × 18 years: DC is in the panel) · treated = pre-2010 ban states · controls = never banned

$$\text{Outcome}_{st} = \beta_1(\text{CU}_t \times \text{CorpBan}_s) + \beta_2(\text{CU}_t \times \text{CorpUnionBan}_s) + \gamma_s + \delta_t + \epsilon_{st}$$

## Two-way fixed effects

Separate treatment effects by pre-existing ban type — the paper’s key refinement over pooled designs.

## Event studies

Leads and lags around 2010 (2009 omitted): pre-trend tests plus the time path of effects.

## Synthetic DiD

Arkhangelsky et al. (2021) reweighting, group-level and state-by-state ATETs; plus a Wooldridge (2025) TWFE check.

Exclusions: CO and SD (bans adopted only in 2002 and 2007), NE (nonpartisan unicameral legislature), LA (nonpartisan blanket primaries). Standard errors clustered by state; coefficients ×100 (percentage points).



# First stage confirmed: legislatures moved right

Replicates Abdul-Razzak et al. (2020) and extends it through the 2022 elections (paper Appendix A)

**Republican vote share:**  $\approx +4\text{--}5\text{pp}$  in ban states after 2010, both chambers (significant with trend controls).

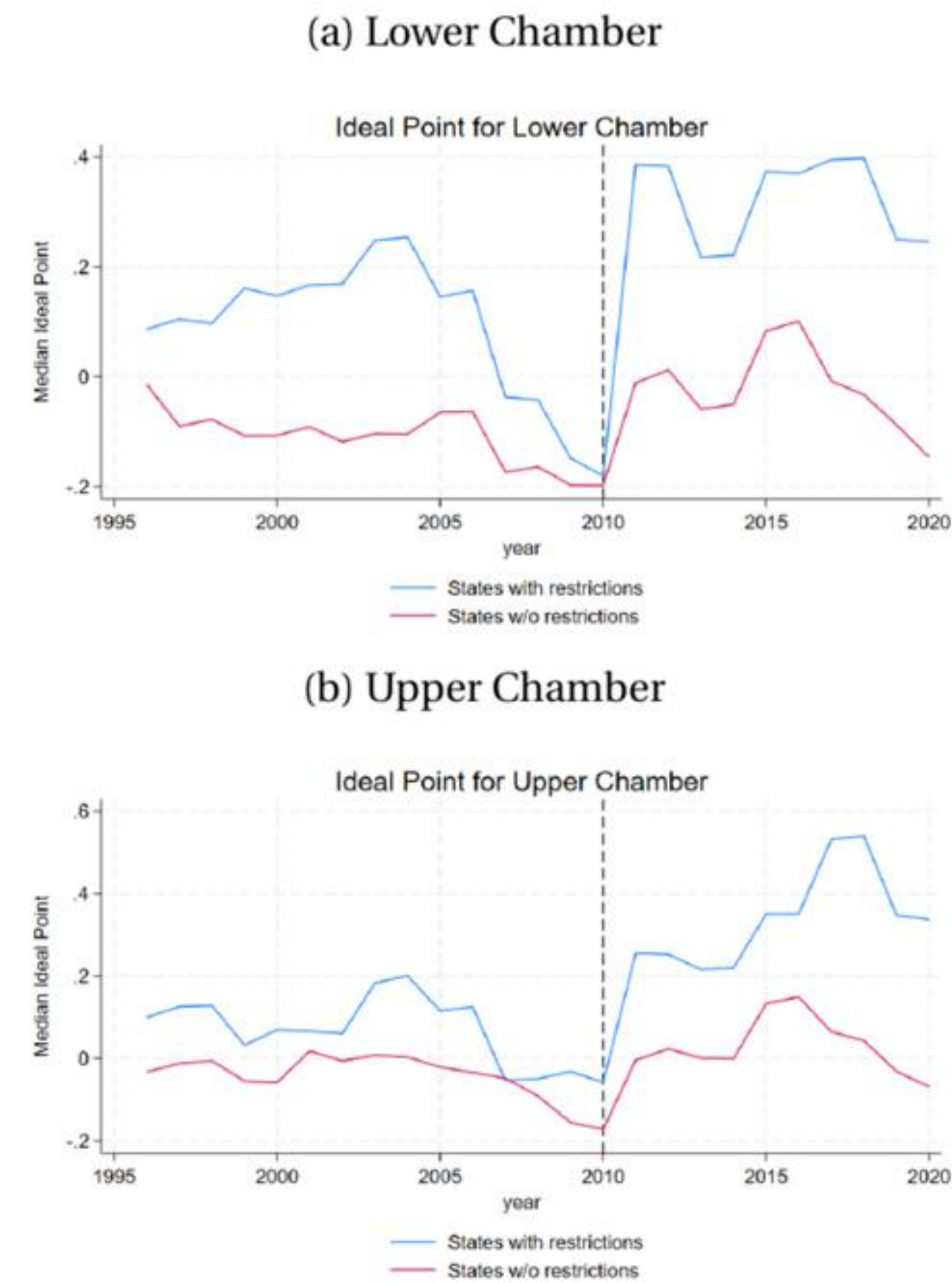
**Median legislator ideology** (Shor–McCarty):  $+0.20$  to  $+0.41$  toward conservative.

**Party control:** Democratic legislative control  $-21\text{pp}^{**}$  and Republican trifectas  $+15\text{pp}^{*}$  — each significant in one of six specifications (Tables A-5, A-7).

Majority parties became more extreme; minority parties moderated. Polarization itself moved little.

By ban type (Table A-4): vote-share effects are not significant in either group, and the ideology shift is significant only in corporate-only-ban states ( $+0.27^{**}$ ).

Figure A–3: Trends in the Median Ideology by Chamber





# Main result: top rates and redistribution fell — only where both bans fell

DiD estimates × 100 (percentage points), state income taxes, 2004–2021 (paper Table 4)

| Citizens United ×              | ATR   | Top 5%  | Top 1%  | Top 5% (wealth) | Top 1% (wealth) | Reynolds–Smolensky |
|--------------------------------|-------|---------|---------|-----------------|-----------------|--------------------|
| Corporate ban only             | 0.01  | 0.14    | 0.30    | 0.07            | 0.10            | 0.07               |
| Corporate & union ban          | −0.01 | −0.33   | −0.53** | −0.25           | −0.36*          | −0.11**            |
| Difference (both – corp. only) | −0.02 | −0.47** | −0.83** | −0.32*          | −0.46*          | −0.18**            |

**Concentrated at the top.** The overall ATR barely moves; top-1% rates fall 0.53pp.

**Asymmetric.** Corporate-only-ban states drift (insignificantly) the other way.

**Redistribution falls.** Reynolds–Smolensky drops where both bans lifted.

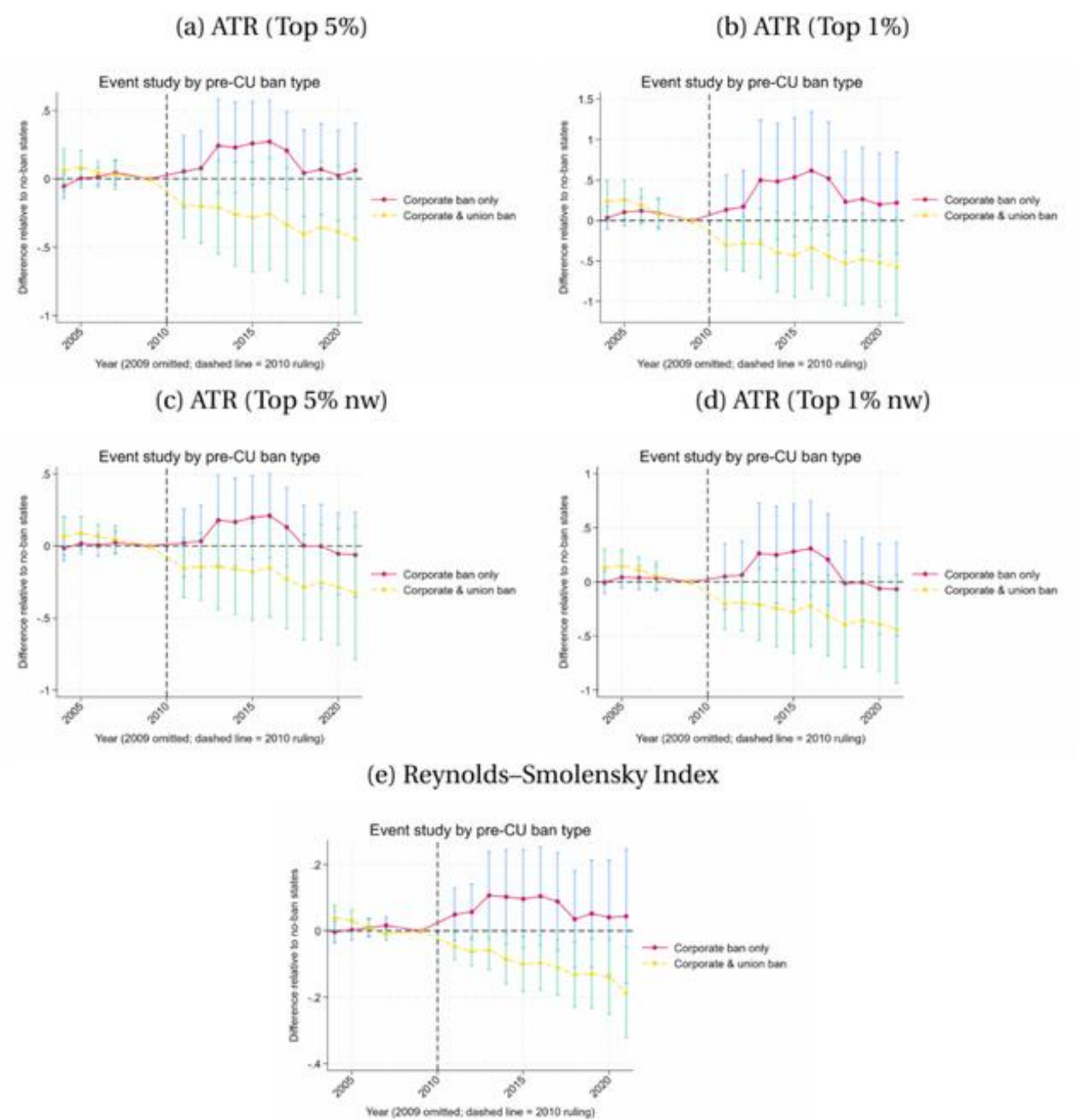
N = 846; SEs clustered by state; \* p<0.10, \*\* p<0.05.



# No significant pre-trends; gradual divergence after 2010

Event studies by pre-existing ban type (paper Figure 3) — joint pre-trend tests pass for every outcome

Figure 3: Event Study Graphs for State-Level Taxes by Ban Type (State Taxes only)



Notes: Each panel reports coefficients from an event-study regression that interacts treatment status by ban type with year-relative-to-2010 indicators, using 2009 as the omitted category. State and year fixed effects are included.

Corporate-and-union-ban states (yellow) drift steadily down; the gap builds over a decade rather than jumping.

Post-2010 coefficients jointly significant for that group: top 1%  $p = 0.025$ , Reynolds–Smolensky  $p = 0.049$  (paper Table 5).

Ban-type differences jointly significant for every outcome except the overall ATR.



# The same signs under stress — with two caveats

Alternative estimators, samples, and income concepts

|                                   |                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Synthetic DiD                     | Same signs; group estimates smaller and not significant (top 1% -0.40, RS -0.07); ban-type differences hold at 10% (-0.72*, -0.13*) |
| Wooldridge (2025) TWFE            | Identical to baseline (top 1% -0.53**, RS -0.11**) — expected with one treatment date and never-treated controls                    |
| Leave-one-out                     | No single treated state drives either group                                                                                         |
| Drop 8 no-income-tax states       | Effects grow (difference: top 1% -1.13***, RS -0.20**) — the authors read the baseline as a lower bound                             |
| After-federal-tax income (App. D) | Larger and sharper (top 1% -0.93***; difference -1.27***, RS -0.24***)                                                              |

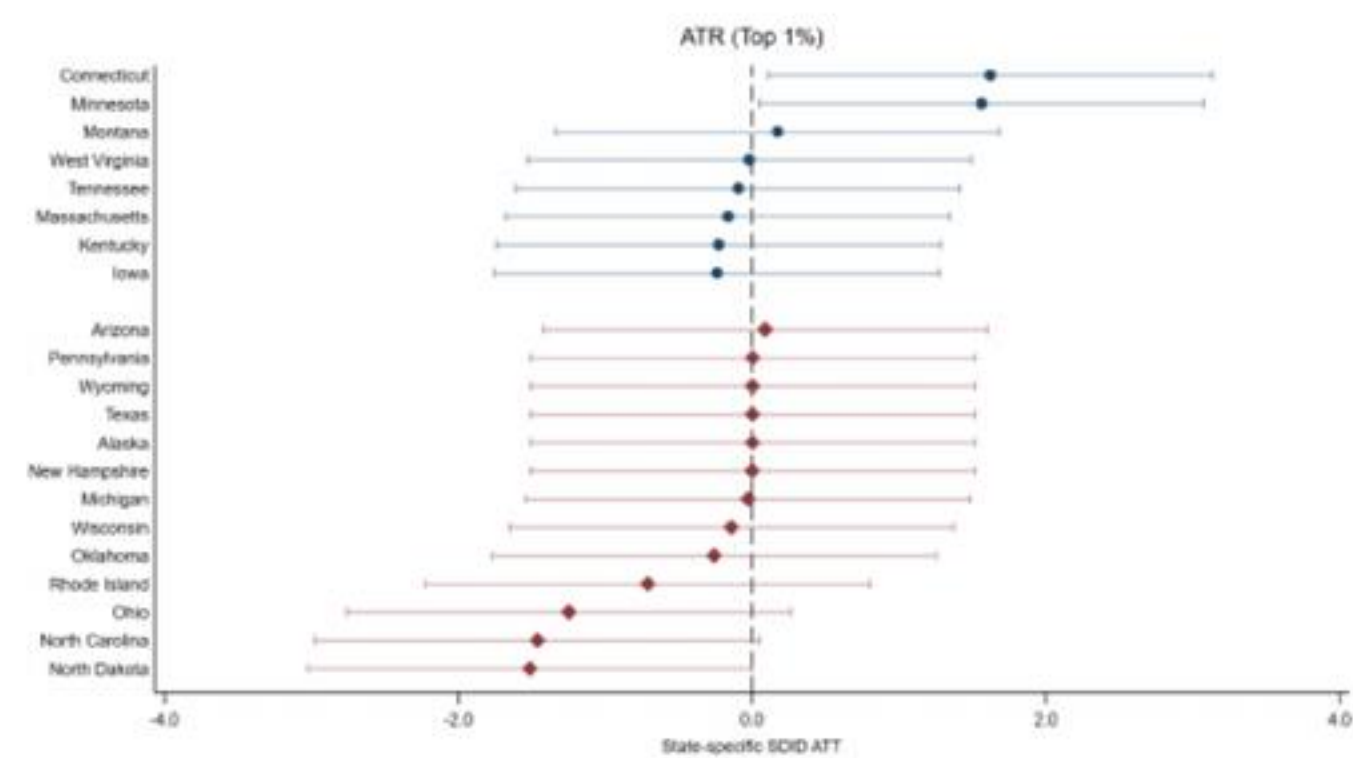
**The honest caveat:** adding state-specific linear trends pulls the corporate-and-union effects “much closer to zero” (Fig. 4), and the SDID group estimates lose significance. The authors treat trends as a robustness check, since they can absorb genuinely gradual treatment effects — exactly the dynamics the event studies show.



# Where the effect lives

State-by-state synthetic DiD effects (paper Figure 5): top-1% ATR and Reynolds–Smolensky

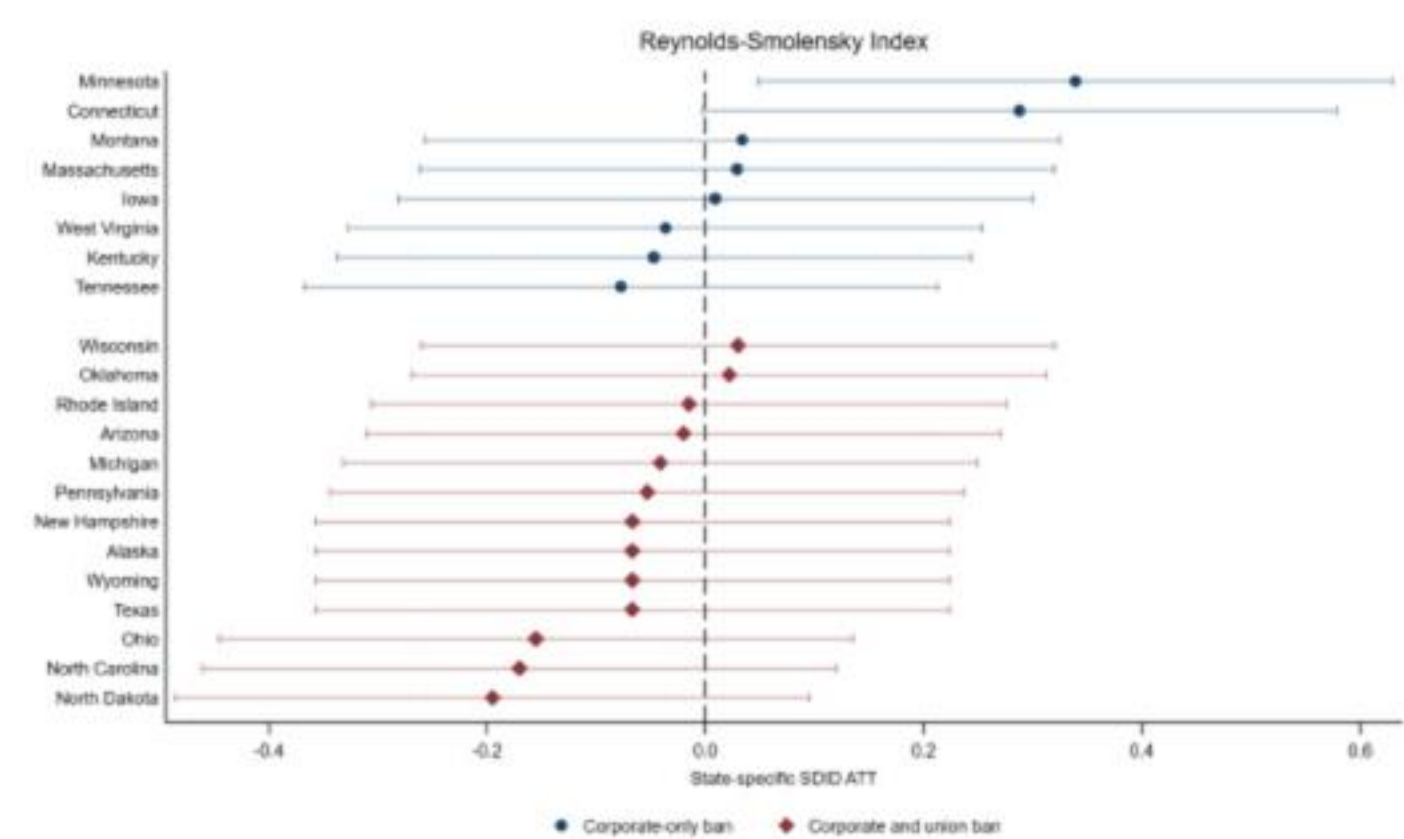
(b) ATR (Top 1%)



**Connecticut and Minnesota** pull the corporate-only group positive — states mostly under Democratic or divided government.

**North Carolina, North Dakota, Ohio** anchor the negative end — Republican-controlled after 2010.

(e) Reynolds–Smolensky Index



The authors say it plainly: “most of our results are driven by few outlying states” — though leave-one-out shows no *single* state is decisive.



# Channels: politics predicts taxes — with a residual

Political outcomes as transmission mechanisms (paper Tables 7, 8, C-2)

## Political composition moves tax outcomes

Republican Senate vote share → redistribution down (RS  $-0.14^*$ ). More conservative median ideology → top-1% ATR  $-0.29^{**}$ . Democratic trifectas and governors → higher top rates and more redistribution; Republican control → the reverse. (These are the significant entries among Table 7's 36 coefficients; polarization is never significant.)

## ...but does not fully explain the ruling's effect

Residualize each tax outcome on vote share, ideology, or polarization, then re-run the DiD: corporate-and-union effects persist (top 1%  $\approx -0.5\text{pp}$ , RS  $\approx -0.1$ , still significant).

Reading: electoral composition carries part of the effect. The remainder points to influence that does not show up in seat counts — **lobbying, donor pressure, agenda-setting**.



# Why the asymmetry? The union counterweight

Union density by pre-existing ban type (paper Figure 6)

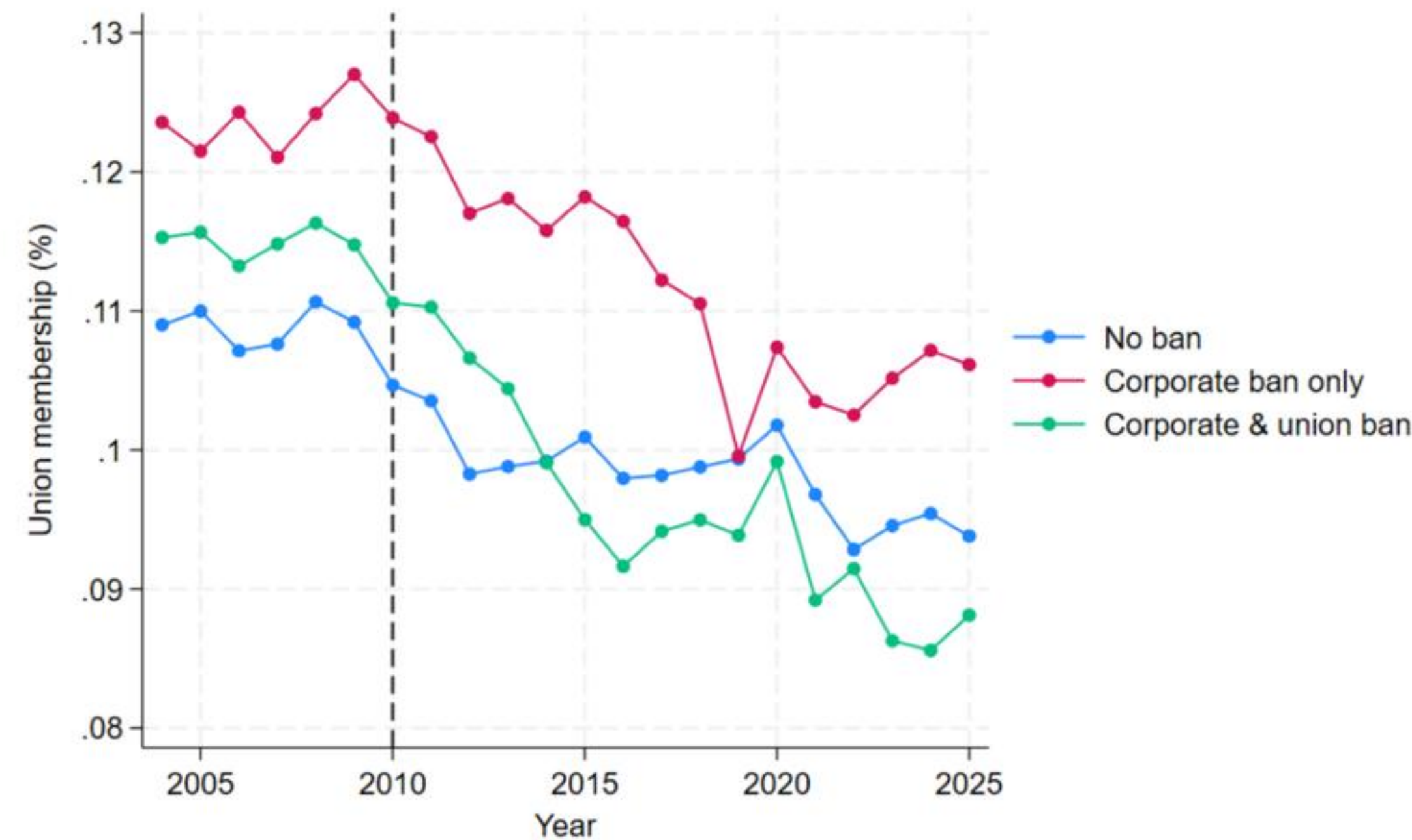
Per the paper (OpenSecrets, 2023–24 federal cycle), unions send  $\approx 87\%$  of their contributions to Democrats; corporate money leans Republican.

**Corporate-only-ban states:** unions could already spend — and are strongest there. New corporate money met an existing counterweight, so competition stayed roughly balanced.

**Corporate-and-union-ban states:** historically Republican-leaning, weaker unions. Lifting both bans favored the side with deeper pockets — the regressive tilt.

In the authors' words: the ruling "amplified pre-existing asymmetries in political influence."

Figure 6: Union Density by Ban Type



Notes: The figure reports average union membership rates, defined as the percentage of employed workers who are union members, computed at the state-year level (Hirsch et al. 2026) and grouped by pre-existing campaign finance regimes across all sectors.

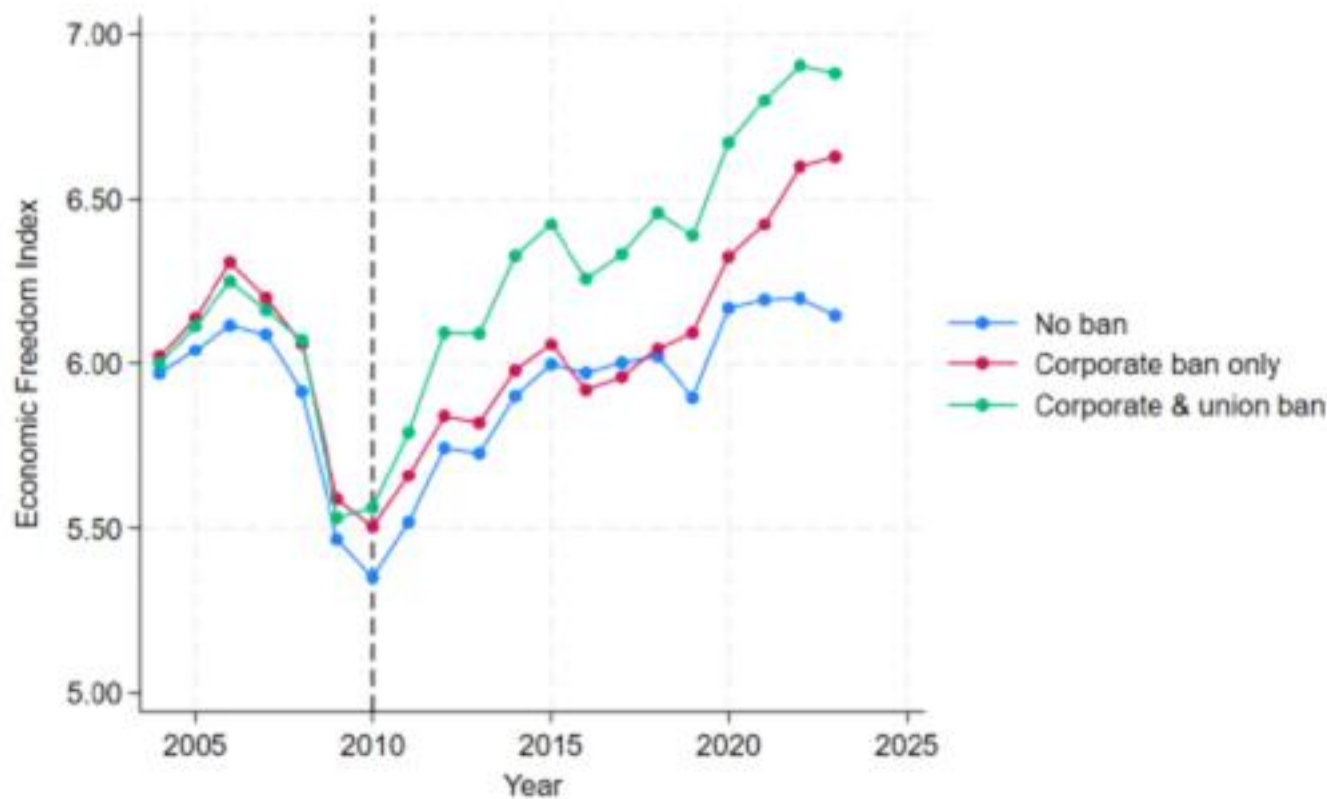


# Part of a broader market-oriented drift

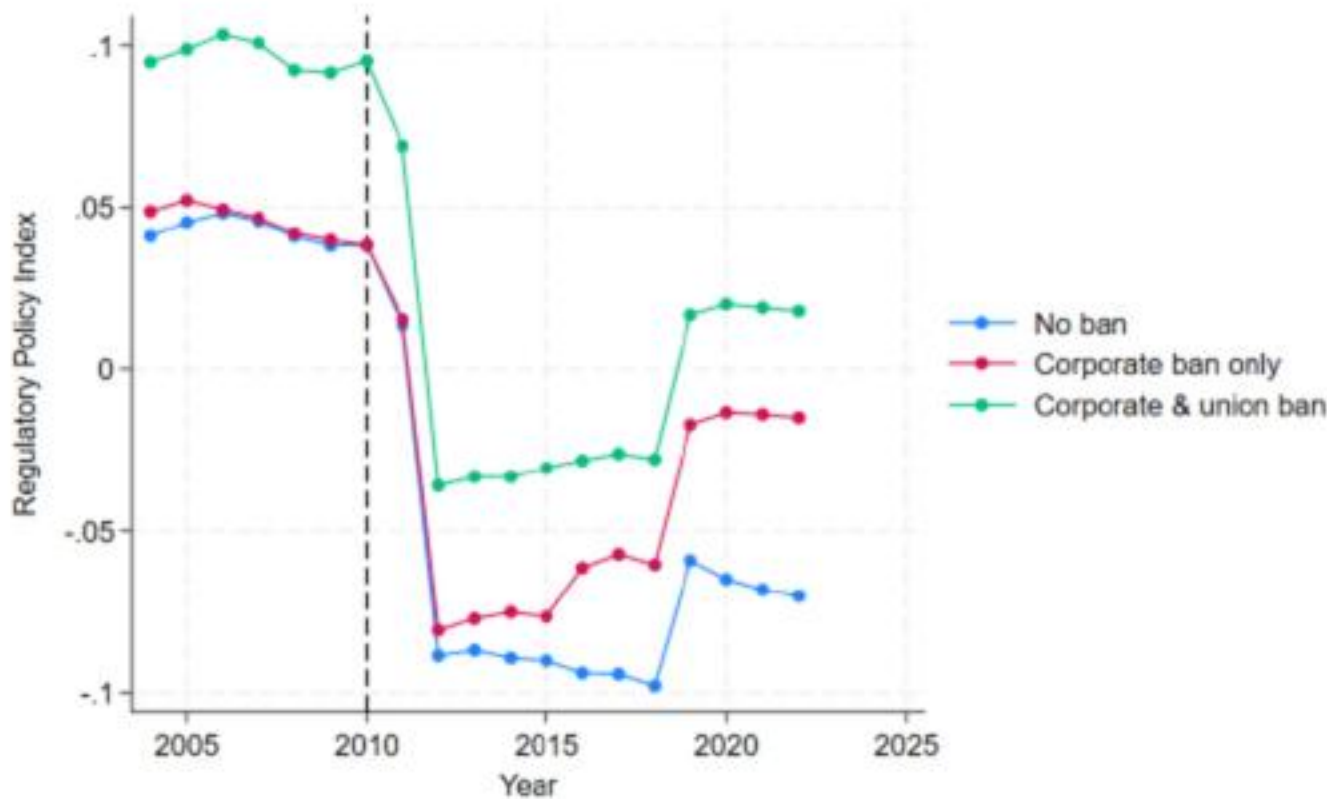
Economic freedom (Fraser) and regulatory policy (Cato) indices by ban type (paper Figure 7)

Figure 7: Business Friendly Environment Indicators

(a) Economic Freedom Index



(b) Regulatory Policy Index



After 2010, both indices rise most in the states that had banned both corporate and union spending — descriptive, and the Fraser index itself includes the top income-tax rate (fn. 9), so panel (a) partly restates the outcome; the Cato regulatory index in panel (b) is the independent check.



# The authors' conclusions

Citizens United changed not only who legislates, but what they legislate

1. Where both corporate and union bans fell, state income taxes became **persistently less progressive**: lower top-group rates, less redistribution, higher post-tax inequality.
2. Effects unfolded **gradually** — policy responds over a decade, not an election cycle.
3. Transmission runs partly through **electoral composition**, partly through influence beyond it.

"Wealthy individuals and corporations can influence policy agendas by financing campaigns and supporting organizations that promote market-oriented and conservative ideologies, resulting in fiscal policies that disproportionately favor their interests ('wingnut welfare')."



# Discussion

Four comments and a public-data replication — from a tax-microsimulation seat



# 1 • The finding is about the union ban — say so up front

Every significant coefficient sits in the 13 states where the *union* ban was also lifted; the abstract describes a general effect

## Table 4, read as a triple difference

Corporate-&-union ban: top 1% **-0.53\*\***, RS **-0.11\*\***. Corporate-only: +0.30, +0.07 (n.s.). The -0.83\*\* difference is, under DiD logic, *the extra effect of also lifting the union ban* — the opposite of what a unions-fund-Democrats story predicts.

## The detectable political first stage is in the corporate-only group (Table A-4)

By ban type, Republican vote share is n.s. in both groups (0.05, 0.01); the ideology shift is significant only in the *corporate-only* states (+0.27\*\*) — where taxes moved up — and the cross-group differences are n.s. Residualizing on vote share or ideology barely moves the tax coefficients (Table 8: -0.47 to -0.56).

The authors' resolution — corporate-&-union states had weaker unions — rests on a ~1pp density gap (Fig. 6). In our public-data replication pre-2010 density is 11.5% vs 12.4% vs 11.5% across the three groups.

## What would convince me

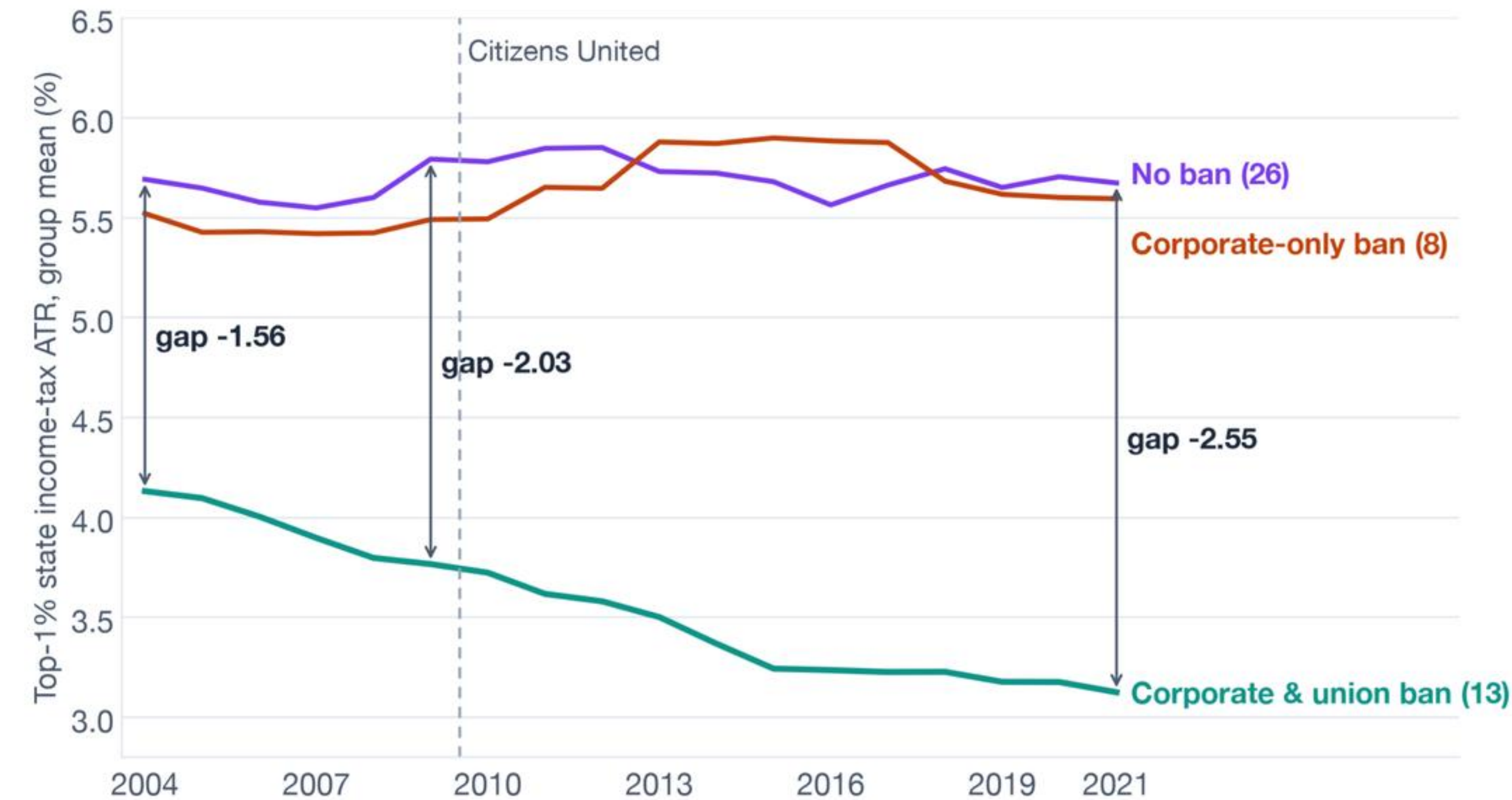
- State the conditionality in the abstract and title the mechanism section after it.
- A dose-response test: CU × pre-2010 union density, within ban groups. In our replication the corporate-&-union effect does *not* scale with density (interaction -0.02, p = 0.57), and CU × density is *positive* — high-union states raised top taxes after 2010.
- We ran the obvious horse race: adding post × Republican-trifecta acquisition (2011–13; 8 of your 13 states, 1 of 8 corporate-only, 6 of 26 controls) moves the coefficient only 16–20% (top 1% -0.57\*, RS -0.07\*); REDMAP targeting, ≈ 0. So the contrast is not just the 2010 wave — which leaves the question sharper: what is it? A theory of change with names would answer it.

Replication: SCF 2010 public extract → TAXSIM-35 (local), 47 units × 18 years; union density from Hirsch–Macpherson (unionstats.com). Everything is open: [github.com/MaxGhenis/iariw-2026-rebechi-replication](https://github.com/MaxGhenis/iariw-2026-rebechi-replication)



## 2 · Which -0.53 do you believe?

The treated-control gap was already widening before 2010 — replication from public data (SCF 2010 → TAXSIM-35, same design), group means by year



Gap -1.56 (2004) → -2.03 (2009) → -2.55 (2021): the pre-period slope is at least as steep as the post-period slope. Joint pre-test  $p = 0.40$  only because clustered SEs are wide.

Consistent with the paper's own checks: state-specific trends pull the effect "much closer to zero" (Fig. 4; -0.27, n.s., in our panel); SDID group estimates are n.s. (Table 6: -0.40, -0.07).

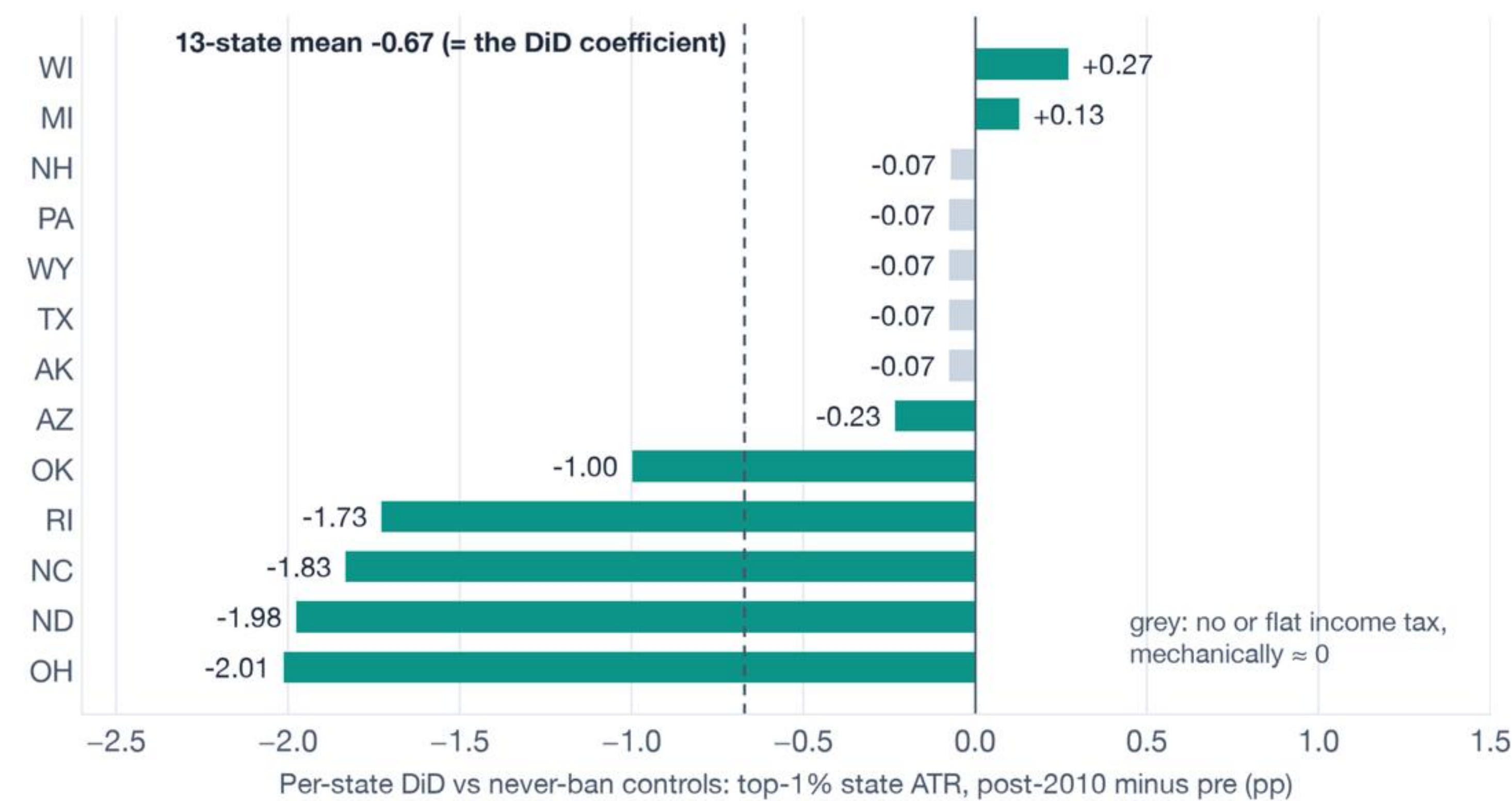
**Asks**  
Report the trends specification as co-equal, and bound the estimate with Rambachan-Roth pre-trend sensitivity.

In fairness: the stars are not a few-cluster artifact. In our replication the top-1% effect survives randomization inference ( $p = 0.013$ ) and a wild cluster bootstrap ( $p = 0.030$ ).



### 3 · Five tax reforms, not thirteen states

Per-state contributions to the corporate-&-union coefficient (replication; the paper's Fig. 5 shows the same ranking with SDID)



Ohio, North Dakota, North Carolina, Rhode Island and Oklahoma carry the estimate. Four treated states have no broad income tax and Pennsylvania's is flat — they cannot respond. Michigan and Wisconsin moved the other way.

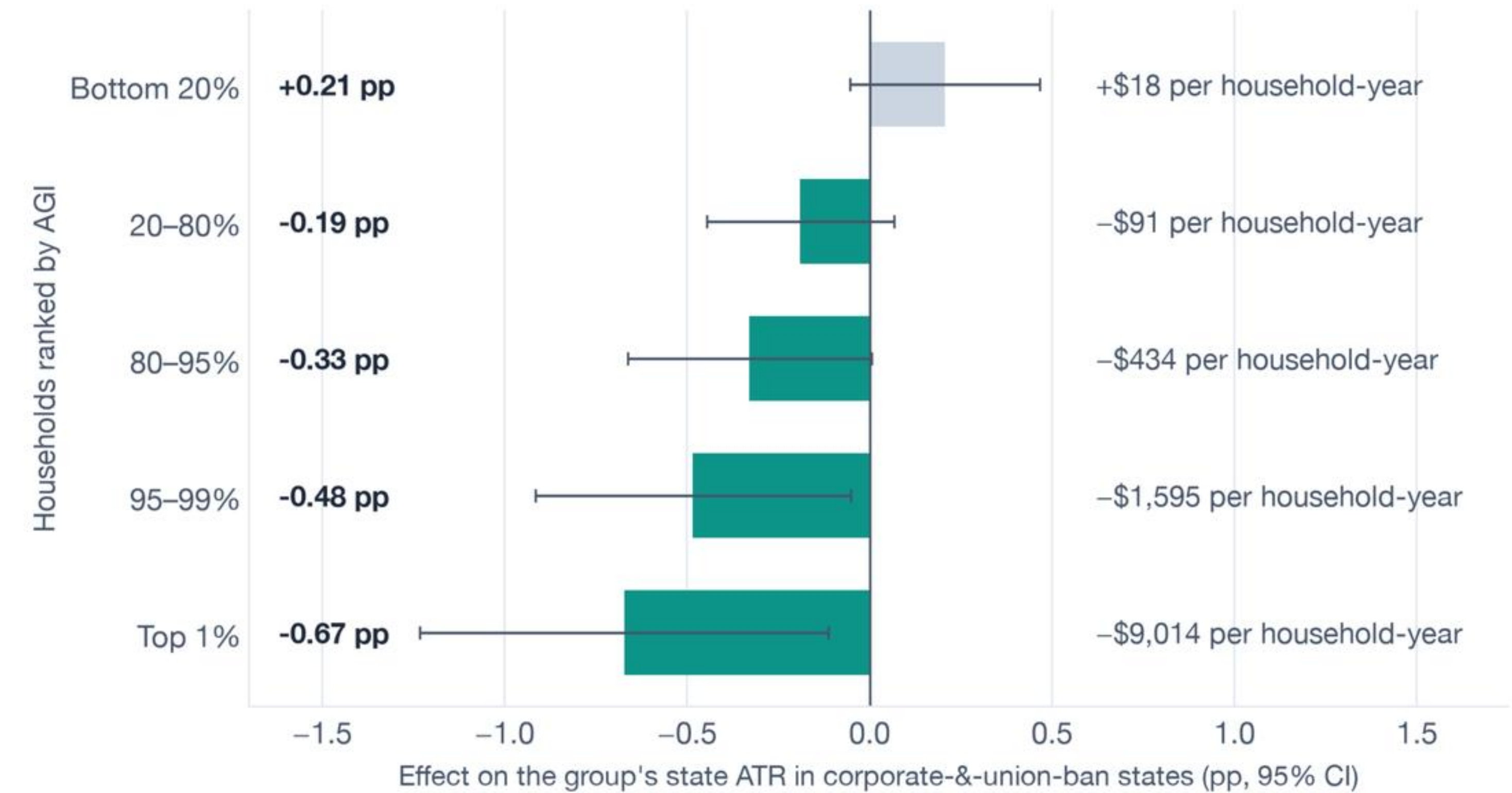
Drop NC, ND and OH together:  $-0.29$  (SE 0.24,  $p = 0.22$ ). Leave-one-out alone cannot show this.

**Asks**  
Consider making the income-tax-states sample (Table B-1) the main one; report leave-three-out; and tell the story of the five reforms — North Dakota's cuts began in May 2009, during the oil boom: a plausible confounder, not a channel.



## 4 · Model the tax function, not six summaries of it

The same 900 simulated datasets support a household-level incidence regression (here in its group-level equivalent) — who got the cut, in dollars (replication, corporate-&-union states)



A monotone gradient: nothing at the bottom, about  $-\$9,000$  a year for the average top-1% household (2010 dollars). Capital-heavy and wage-heavy top households are hit alike — across-the-board top-rate cuts, not capital carve-outs.

With a fixed sample, the saturated micro regression *is* the paper's  $\Phi$  by subgroup — so this costs nothing new and turns "RS  $-0.11$ " into an incidence table. And contrasts difference out common schedule shocks: the  $\beta$  the paper defines in eq. (5) but never reports —  $ATR(\text{top } 20\%) - ATR(\text{bottom } 20\%)$  — is its most precise outcome in our replication:  $-0.68$  (SE 0.22,  $t = 3.1$ ).

**Asks**  
Report  $\beta$  (eq. 5, defined but never shown) and the dollar incidence; say that RS  $-0.11$  is  $-0.0011$  Gini points on a state-tax redistribution of about 0.004 — and say whether benefits B (eq. 1) enter the simulation at all.



# The headline reproduces from public data

SCF 2010 public extract → NBER TAXSIM-35 (run locally) → 51 jurisdictions × 18 years × 6,482 households → eq. (7). Coefficients × 100, SEs clustered by state

| Table 4 row                                | Paper          | Replication    |
|--------------------------------------------|----------------|----------------|
| Top-1% ATR, corporate & union ban          | −0.53** (0.24) | −0.67** (0.29) |
| Reynolds–Smolensky, corporate & union ban  | −0.11** (0.04) | −0.09** (0.03) |
| Difference (both – corporate only), top 1% | −0.83** (0.35) | −0.87** (0.38) |
| Corporate-only arm                         | positive, n.s. | positive, n.s. |
| Income-tax states only (Table B-1), top 1% | −0.77** (0.30) | −0.95** (0.36) |
| Overall ATR, corporate & union ban         | −0.01 (0.13)   | −0.35** (0.17) |

N = 846 only with DC in the panel (47 × 18) — the text says 46 states. “SCF 2011” is the 2010 wave (LWS US10; income year 2009): the SCF is triennial.

What does not reproduce: the clean zero on the overall ATR. The public extract forces no itemized deductions and taxes Social Security as pensions, which inflates middle-class liabilities. The top-1% and RS rows are insensitive to this.

Full pipeline ≈ 25 minutes of compute; one SCF implicate; TAXSIM vintage differs. Scripts, panel and results: [github.com/MaxGhenis/iariw-2026-rebechi-replication](https://github.com/MaxGhenis/iariw-2026-rebechi-replication). A released 47 × 18 panel plus code would make the core table reproducible in an afternoon.



# Where this sits in the Citizens United evidence

What the cited and uncited empirical work actually finds — every entry retrieved and checked

|                                                                                           |                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gilens, Patterson &amp; Haines (2021, APSR) — not cited</b>                            | Same natural experiment, generalized synthetic control: top corporate tax $-2.83$ ( $p = 0.01$ ; 0–100 index), and twice as large in the eight corporate-only states ( $-5.55$ , vs all treated) — the union counterweight hypothesized ex ante, in the mirror image of this paper.             |
| <b>Slattery, Tazhitdinova &amp; Robinson (2023, JPubE) — cited as “a moderate effect”</b> | A precise null on top corporate, top personal-income and sales rates, incentives and revenues (top PIT $\approx -0.46$ pp, n.s.); corporate-only restriction: a tighter null. Parallel trends fail in levels, so they use logs.                                                                 |
| <b>Klumpp, Mialon &amp; Williams (2016, JLE)</b>                                          | CU raised Republican House-race wins $\approx 4$ pp; descriptively largest in MN, MT, MI, OH, IA — three of them corporate-only states (no ban-type test reported). The corporate-funded 2010 REDMAP drive targeted MI, OH, PA, TX, NC, WI for redistricting, not for ban type.                 |
| <b>Abdul-Razzak, Prato &amp; Wolton (2020, Electoral Studies)</b>                         | Vote-share effect $\approx 4$ – $5$ pp (trend-dependent); $\approx 8$ pp where unions are weak, $\approx 0$ where strong — by density, not ban type. In treated disclosure states Democratic-aligned groups still outspent Republican-aligned ones after CU; fn. 12: no effect on public goods. |
| <b>Hansen, Rocca &amp; Ortiz (2015, JOP); Spencer &amp; Wood (2014)</b>                   | Major corporations “were not a source of the dramatic increase in independent spending” (federal, 2012); the state-level surge ran through 501(c)/527 vehicles whose donors are unobservable — $\approx \$1.6$ M per treated state over 2010–12.                                                |
| <b>Akey et al. (2023, NBER); Werner &amp; Coleman (2015, JLEO); Farver (2024)</b>         | The policy record is mixed, not “mostly null”: enforcement and antitakeover effects exist; tax rates negative but n.s. (PIT $\approx -0.3$ pp); no effect on environmental policy or overall policy liberalism; turnover rose in both parties.                                                  |



# Questions to open the floor

**Federal analogue.** There is no US-without-Citizens-United counterfactual at the federal level. Do these state results bound what happened in Congress?

**The 2021–2023 tax-cut wave.** Many corporate-and-union-ban states cut income taxes after the sample ends in 2021. Does extending the panel sharpen the effect?

**Who actually spent?** Your Table 1 shows the two sides near parity by 2020–24 at the federal level. What does the state-level ledger look like in the 13 states that drive the result?

---

Thank you — and over to **Philippe Van Kerm**

Max Ghenis · [max@policyengine.org](mailto:max@policyengine.org) · slides, memo, replication: [github.com/MaxGhenis/iariw-2026-rebechi-replication](https://github.com/MaxGhenis/iariw-2026-rebechi-replication)